

Program: Diploma in Electrical Engineering & Electric Vehicle Technology	
Course Code: 4411	Course Title: Vehicle Chassis and Transmission Systems
Semester: 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To assimilate knowledge about the operation of various parts of driveline, suspension, steering and braking system of vehicles.
- To acquire the knowledge about fundamentals regarding automobile driveline, suspension, steering and braking systems along with modern features.
- To enable the student to perform trouble shooting and servicing of modern vehicles.

Course Prerequisites:

Topic	Course code	CourseTitle	Semester
Basic Knowledge about vehicles and chassis	2411	Introduction to Electric Vehicles & Elementary Theory.	2
Knowledge about Force Analysis	1003, 2003	Applied Physics-I & II	1, 2

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive level
CO1	Describe the working of various types of clutches And transmission system	15	Understanding
CO2	Understand the features of driveline components	14	Understanding
CO3	Describe the various suspension and steering systems	15	Understanding
CO4	Explain the working construction of modern Brake systems.	14	Understanding
	SeriesTest	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Describe the working of various types of clutches and transmission system		
M1.01	Describe the various clutch components and their functions	4	Understanding
M1.02	Outline various clutch linkages	3	Understanding
M1.03	Explain the purpose and operation of manual Transmission and transaxle	4	Understanding
M1.04	Explain the operation of automatic transmission	4	Understanding
Contents: Functions of clutch, clutch operation, types of clutches, clutch components-clutch disc, throw out bearing, pressure plate assembly, clutch pedal linkage-levers and rods, cable, hydraulic, dual mass flywheel. Need for transmission, transmission vs transaxle, transmission design, transaxle design, synchronizers, gear shift mechanisms, transmission, and transaxle power flows. Automatic transmission-torque converter, shift modes, planetary gear sets, planetary gear controls, hydraulic systems, dual clutch automatic transmission.			
CO2	Understand the features of driveline components		
M2.01	Identify the various driveline components	4	Understanding
M2.02	Explain the working of differential	3	Understanding
M2.03	Identify the features of wheels and tyres	3	Understanding
M2.04	Outline the factors in wheel alignment geometry	4	Understanding
	Series Test I	1	

Contents:

Driveline, universal joint operation, constant velocity joints-purpose and function, outer and inner cv joints, drive axle shafts, differential-parts and operation, limited slip differential, four-wheel drive systems, all-wheel drive, transfer case, inter axle differential.

Wheels-types, construction, wheel dimensions, tyres-types, construction, tread designs, tyre ratings, tyre designations, tyre care, tyre pressure monitor, tyre repair, wheel alignment geometry-caster, camber, toe, steering axis inclination, turning radius.

CO3	Describe the various suspension types and steering systems		
M3.01	Identify various suspension components and identify their features	4	Understanding
M3.02	Outline the features of front and rear Suspension types	4	Understanding
M3.03	Identify various steering components.	4	Applying
M3.04	Explain the working of various power steering systems	3	Understanding

Contents:

Suspension system-purpose and function, suspension principles, suspension parts-steering knuckle, control arms, ball joints, strut rods, stabilizer bars, front suspension types-solid axle, rear suspension types-solid axle, leaf spring, trailing arm, independent rear suspension, electronic suspension-types, air suspension-components, working.

Steering linkages-parallelgram steering linkage-components, rack and pinion steering linkage-components, steering gears-recirculating ball, worm and roller, tilt steering wheel, collapsible steering column, power steering systems-integral piston system, power assisted rack and pinion systems, electronic power steering systems.

CO4	Explain the working construction of modern brake systems.		
M4.01	Identify the components in hydraulic brake system	4	Understanding
M4.02	Explain the working of master cylinder and wheel cylinder	3	Understanding
M4.03	Explain the features of drum brake and disc brake	4	Understanding
M4.04	Summarize the working of power assisted modern brake systems	3	, Understanding
	Series Test II	1	

Contents:

Factors governing braking, hydraulic brake system -components, brake pedal, dual piston master cylinders-construction and operation, wheel cylinder-construction and operation, split hydraulic systems, hydraulic lines, pressure differential switch, metering, and proportionating valves.

Drumbrake-componentsandfunction, discbrake-componentsandfunction, types of caliper assembly, power assisted brakes, vacuum brake booster, hydraulic brake booster, air brake-components, working of ABS braking system.

Text/Reference:

T/R	Book Title/Author
R1	Automotive Technology: Principles, Diagnosis, and Service by James D. Halderman, Prentice Hall
R2	Automotive Technology: A Systems Approach by Jack Erjavec, Cengage Learning
R3	P.L.Kohli - Automobile Chassis & body –Tata McGraw-hill
R4	Kripal Singh-Automobile Engg Vol-1 Standard Publication
R5	WH Crouse – Automotive Chassis, McGraw - Hill Book Company
R6	Anil Chhikara-Automobile Engineering –Satya Prakasan

OnlineResources:

Sl.No	WebsiteLink
1	https://www.youtube.com/watch?v=devo3kdSPQY
2	https://www.youtube.com/watch?v=u_y1S8C0Hmc
3	https://www.youtube.com/watch?v=SOgoejxzF8c
4	https://www.youtube.com/watch?v=bMg_j5_AGMg